

The rules of differentiation

Power, constant multiple, sum, product and quotient — five rules that replace first principles for good.

LESSON 7–9

3 × 40 MIN

ALGEBRA 11, §1.4

P1 · 7.2–7.4

LESSON CLOCK

14'

16'

26'

26'

32'

21'

The power rule	14 min
Sums and multiples	16 min
The product rule	26 min
The quotient rule	26 min
Practice	32 min
Homework and consolidation	21 min

LEARNING OBJECTIVES

- Apply the power rule to positive, negative and fractional indices.
- Differentiate sums and constant multiples.
- Apply the product rule and the quotient rule.
- Rewrite an expression into a differentiable form before starting.

TEXTBOOK

National	pp. 39–54
Cambridge	Pure Mathematics 1, pp. 130–145
Workbook	P1 Exercise 7B–7D

01

Explanation

The power rule

$$(x^n)' = n \cdot x^{n-1} \text{ for every real } n$$

$F(X)$	REWRITE AS	$F'(X)$
x^5	x^5	$5x^4$
$\sqrt{x}$	$x^{1/2}$	$\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$
$\frac{1}{x^3}$	x^{-3}	$-3x^{-4} = -\frac{3}{x^4}$
$\frac{1}{\sqrt{x}}$	$x^{-1/2}$	$-\frac{1}{2x\sqrt{x}}$
x	x^1	1
7	$7x^0$	0

REWRITE BEFORE YOU DIFFERENTIATE

Every root becomes a fractional index; every fraction with x alone below becomes a negative index. Do that first, and the power rule handles everything.

Multiples and sums

$$(c \cdot f)' = c \cdot f' \quad (f \pm g)' = f' \pm g'$$

Together these differentiate any polynomial term by term:

$$(4x^3 - 7x^2 + 2x - 9)' = 12x^2 - 14x + 2$$

THERE IS NO PRODUCT RULE FOR A CONSTANT

$(5x^3)'$ is $15x^2$, not $(5)'(x^3) + 5(x^3)'$. Constants pass straight through — using the product rule here is not wrong, just wasteful.

The product rule

$$(u \cdot v)' = u'v + uv'$$

THE DERIVATIVE OF A PRODUCT IS NOT THE PRODUCT OF THE DERIVATIVES

Test it: $(x \cdot x)' = (x^2)' = 2x$, but $x' \cdot x' = 1$. The rule is needed.

Method. Write u and v in a column, differentiate each beside it, then assemble. For $y = x^3(2x + 1)$:

	U	V
function	x^3	$2x + 1$
derivative	$3x^2$	2

$$y' = 3x^2(2x + 1) + x^3 \cdot 2 = 6x^3 + 3x^2 + 2x^3 = 8x^3 + 3x^2$$

Check by expanding first: $y = 2x^4 + x^3$, so $y' = 8x^3 + 3x^2$. ✓

The quotient rule

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

THE ORDER IN THE NUMERATOR MATTERS

It is $u'v$ **minus** uv' , not the other way round. Swapping them flips the sign of every answer. The product rule is symmetric; the quotient rule is not.

For $y = \frac{x^2}{x + 1}$:

$$y' = \frac{2x(x + 1) - x^2 \cdot 1}{(x + 1)^2} = \frac{x^2 + 2x}{(x + 1)^2}$$

OFTEN YOU DO NOT NEED IT

$y = \frac{x^3 + 2x}{x}$ is just $x^2 + 2$. Divide through first whenever the denominator is a single term — it is faster and safer.

02 Worked examples

EXAMPLE 1 Differentiate $y = 3x^4 - \frac{2}{x^2} + 5\sqrt{x}$.

1 Rewrite: $3x^4 - 2x^{-2} + 5x^{1/2}$.

2 $12x^3$

3 $+ 4x^{-3}$

Note the double negative.

4 $+ \frac{5}{2}x^{-1/2}$

ANSWER

$$y' = 12x^3 + \frac{4}{x^3} + \frac{5}{2\sqrt{x}}$$

EXAMPLE 2 Differentiate $y = (x^2 + 1)(3x - 4)$ by the product rule.

1 $u = x^2 + 1, u' = 2x$

2 $v = 3x - 4, v' = 3$

3 $y' = 2x(3x - 4) + (x^2 + 1)(3)$

4 $= 6x^2 - 8x + 3x^2 + 3 = 9x^2 - 8x + 3$

ANSWER

$$9x^2 - 8x + 3$$

EXAMPLE 3 Differentiate $y = \frac{2x - 1}{x^2 + 3}$.

$$① \quad u = 2x - 1, u' = 2$$

$$② \quad v = x^2 + 3, v' = 2x$$

$$③ \quad y' = \frac{2(x^2 + 3) - (2x - 1)(2x)}{(x^2 + 3)^2}$$

$$④ \quad = \frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}$$

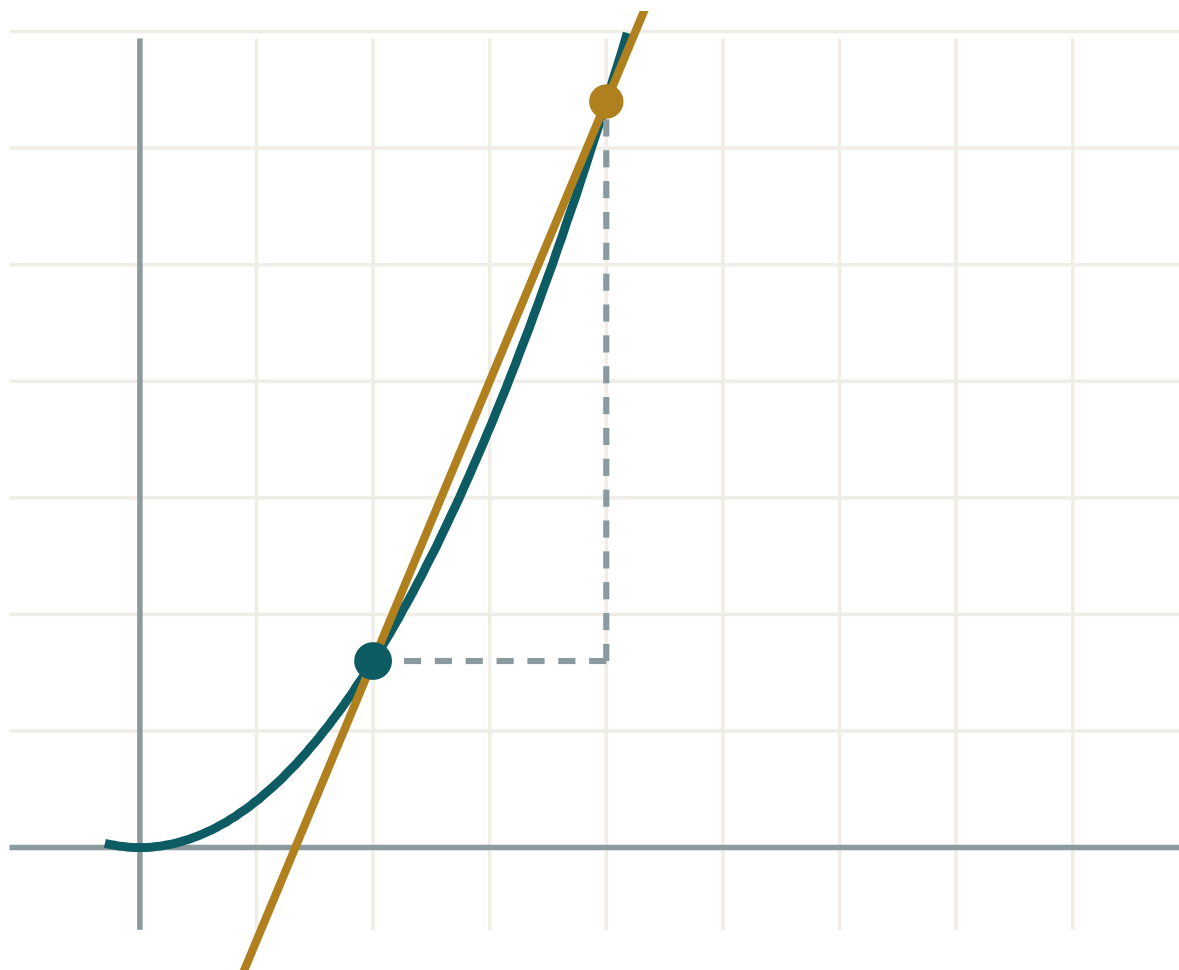
ANSWER

$$\frac{-2x^2 + 2x + 6}{(x^2 + 3)^2}$$

03 Interactive model

Do one product both ways — expand first, then the rule — and confirm they agree.

Checking a rule


 x 2

 h 2

 $\Delta y / \Delta x$ 2.4

 $f'(x)$ exactly 1.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Power rule	Daraja qoidasi	Правило степени
Constant multiple rule	O'zgarmas ko'paytuvchi qoidasi	Правило постоянного множителя
Sum rule	Yig'indi qoidasi	Правило суммы
Product rule	Ko'paytma qoidasi	Правило произведения
Quotient rule	Bo'linma qoidasi	Правило частного
Fractional index	Kasr ko'rsatkich	Дробный показатель
Negative index	Manfiy ko'rsatkich	Отрицательный показатель
Simplify first	Avval soddalashtirish	Сначала упростить
Second derivative	Ikkinchi tartibli hosila	Вторая производная

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself $(x^7)'$ is:

$7x^6$

$7x^8$

x^6

$6x^7$

 $(\sqrt{x})'$ is:

$\frac{1}{2\sqrt{x}}$

$2\sqrt{x}$

$\frac{1}{\sqrt{x}}$

$\frac{1}{2}$

The product rule is:

$u'v'$

$u'v + uv'$

$u'v - uv'$

$uv' - u'v$

The quotient rule numerator is:

$$u'v + uv'$$

$$u'v - uv'$$

$$uv' - u'v$$

$$u'v'$$

$(\frac{1}{x^4})'$ is:

$$-\frac{4}{x^5}$$

$$\frac{4}{x^5}$$

$$-\frac{1}{x^5}$$

$$-4x^3$$

$(6)'$ is:

$$6$$

$$1$$

$$0$$

$$6x$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(x^6)'$

ANSWER $6x^5$

2. $(3x^2)'$

ANSWER $6x$

3. $(x^3 + x)'$

ANSWER $3x^2 + 1$

4. $(5)'$

ANSWER 0

5. $(4x)'$

ANSWER 4

6. $(\frac{1}{x})'$

ANSWER $-\frac{1}{x^2}$

7. $(\sqrt{x})'$

ANSWER $\frac{1}{2\sqrt{x}}$

Medium · the standard question

7 PROBLEMS

1. $(2x^4 - 3x^2 + 7)'$

ANSWER $8x^3 - 6x$

2. $(\frac{3}{x^2})'$

ANSWER $-\frac{6}{x^3}$

3. $(x\sqrt{x})'$

ANSWER $\frac{3}{2}\sqrt{x}$

4. $((x+1)(x-3))'$

ANSWER $2x - 2$

5. $(x^2(x+5))'$

ANSWER $3x^2 + 10x$

6. $(\frac{x}{x+1})'$

ANSWER $\frac{1}{(x+1)^2}$

7. $(\frac{x^2+1}{x})'$

ANSWER $1 - \frac{1}{x^2}$

Hard · stretch and reason**7 PROBLEMS**

1. $((2x - 1)(x^2 + 3))'$

ANSWER $6x^2 - 2x + 6$

2. $(\frac{2x + 3}{3x - 1})'$

ANSWER $-\frac{11}{(3x - 1)^2}$

3. $(x^3\sqrt{x})'$

ANSWER $\frac{7}{2}x^{(5/2)}$

4. $(\frac{\sqrt{x}}{x + 1})'$

ANSWER $\frac{1 - x}{2\sqrt{x}(x + 1)^2}$

5. $(x^2(x - 1)(x + 1))'$

ANSWER $4x^3 - 2x$

6. Find y'' for $y = x^4 - 3x^2$

ANSWER $12x^2 - 6$

7. Find x where $y = x^3 - 3x$ has gradient 9

ANSWER $x = \pm 2$

07 Homework**Homework — 6 tasks**

Rewrite roots and fractions as indices before differentiating.

1. Differentiate $y = 5x^4 - \frac{3}{x} + 2\sqrt{x}$.

2. Differentiate $y = (3x + 2)(x^2 - 1)$ twice — by expanding and by the product rule.

3. Differentiate $y = \frac{x-2}{x+2}$.

4. Differentiate $y = x^2\sqrt{x}$.

5. Find y'' for $y = 2x^5 - x^3 + 4x$.

6. Find the points on $y = x^3 - 12x$ where the tangent is horizontal.